

TIC 43330648

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 14.285582 d
Epoch (t0) 1991.2750 BTJD
Duration 0.1700 d
Depth 0.031 % (312 ppm)
SNR 15.1

CATALOG CROSS-MATCH (novelty)

VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 6 (gate >= 3)
Robust depth ratio 0.94 (gate >= 0.50)
Robust depth sigma 10.9 (gate >= 7.0)
Odd/even z-score 0.62 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 26 light-curve dip in this sector: 6.5 sigma t0 refined by 0.1325 phase
Difference-image peak SNR 25.82 (gate >= 5.0) Compactness 0.84 (gate >= 0.50)
Dimming centroid offset 0.79 px (17 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 43330659
2.01 px away, dTmag +7.83 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

